INFORMATION SECURITY | PROJECT 1 - PCAP ANALYSIS

PCAP to SQLite Network Traffic Analyzer

Project Report

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1 Executive Summary

The PCAP to SQLite Network Traffic Analyzer is an offline tool that converts a recorded packet capture into a searchable SQLite database and presents the result through a local browser interface. It reads an authorized PCAP file, extracts packet metadata (time, addresses, protocol, ports and length), stores one record per packet, and lets the user review summaries, apply filters and export selected rows to CSV.

The workflow runs from file selection through validation, import, traffic review, filtering and export. The PCAP to SQLite conversion is the core implemented module of the project. This report describes the scope, design, data model, interface, validation method and safeguards of the analyzer.

2 Project Overview

2.1 Problem statement

Raw PCAP files are valuable evidence, but repeated manual filtering is slow when a student or analyst needs to compare hosts, protocols, ports and time periods. Storing the extracted metadata in a local database makes it easier to search and reuse, while each row keeps a link back to the original packet number.

2.2 Objectives

- Import an authorized PCAP file and validate its format before processing.
- Parse packet time, addresses, transport protocol, ports, packet length and a compact summary where available.
- Insert the extracted metadata into an indexed SQLite database together with the capture source and import status.
- Provide filters and simple counts so a user can inspect traffic without writing SQL.
- Export filtered metadata to CSV for coursework review.

2.3 Technology stack

Technology	Role
Python	Import orchestration and analysis logic
Scapy	Offline reading and decoding of PCAP packets
SQLite (sqlite3)	Local structured storage and queries
Streamlit	Local browser interface for import, filtering and display
CSV	Export of selected metadata

3 Scope and Methodology

3.1 Scope

Included	Excluded
Offline import of .pcap files	Live packet capture or network scanning
IPv4 and IPv6 packet metadata where decoded	Payload inspection and file reconstruction
TCP, UDP, ICMP and other protocol labels	Intrusion detection or automatic threat verdicts
Capture summaries, filters and CSV export	Multi-user accounts or cloud hosting
SQLite storage with indexed search fields	Full PCAP reconstruction from the database

3.2 End-to-end method

1. The user chooses a locally stored, authorized PCAP file.
2. The application checks the extension, the size limit and whether the parser can open the capture.
3. The parser reads packets one at a time and normalizes supported metadata; missing fields remain null.
4. The importer writes rows to SQLite in batches and records the capture hash and import status.
5. The interface queries the database for summaries and filtered rows; exports contain only the selected metadata.

3.3 Assumptions and limits

The system is designed for a single local user, reasonably sized classroom captures and authorized inspection of the supplied traffic. Some packets have no IP addresses or transport ports, so the database permits missing values. Encrypted application traffic can be counted and classified by its visible metadata, but its content cannot be inferred. The analyzer reports observations only and does not identify attacks on its own.

4 System Design

The system separates file handling, packet decoding, storage and presentation. This allows the PCAP to SQLite module to be verified independently of the interface.

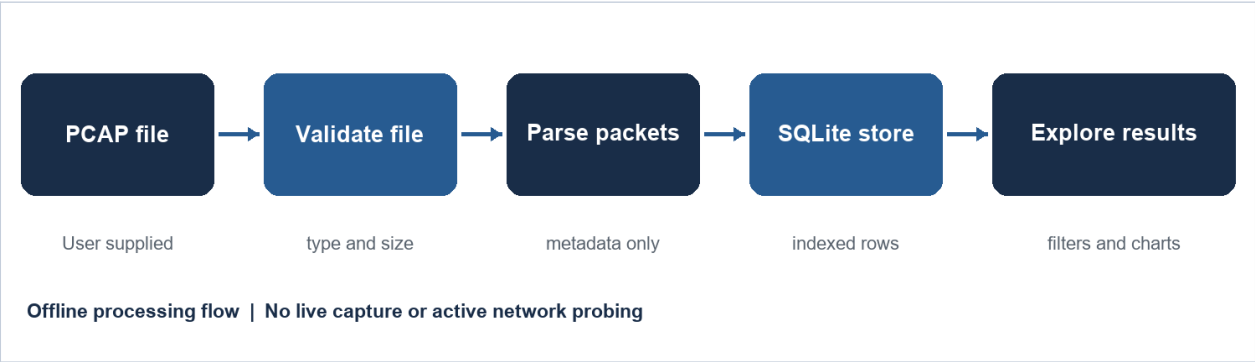


Figure 1: Offline processing flow (no live capture or active network probing)

4.1 Module responsibilities

Module	Responsibility	Output
File intake	Validate the file and create a capture record	Capture ID and status
Packet parser	Decode supported packet layers	Normalized packet fields
Database writer	Batch-insert rows and build indexes	SQLite tables
Analysis view	Run fixed, parameterized queries	Counts and filtered rows
Export	Write selected result rows	CSV file

4.2 Processing rules

- Each imported packet keeps its original sequence number so results can be checked against the source capture.
- Timestamps are stored in a consistent numeric form and displayed in readable local time.
- Packet parsing errors are counted, and a file-level failure marks the import as incomplete.
- All filter queries use SQL parameters.

5 Data Model

The database uses a **captures** table to track imports and a **packets** table to store extracted metadata. One capture can contain many packet rows. Only metadata is stored, not raw payloads, which reduces database size and limits exposure of sensitive content.

Table	Field(s)	Purpose
captures	capture_id	Primary key
captures	filename, sha256	Source identity and duplicate check
captures	imported_at, status	Audit of import progress
captures	packet_count, error_count	Completion summary
packets	packet_id, capture_id	Row key and parent capture
packets	packet_number, timestamp	Traceability and time filter
packets	src_ip, dst_ip	Endpoint filtering
packets	protocol, src_port, dst_port	Protocol and port analysis
packets	length_bytes, summary	Size and short packet description

5.1 Queries and indexes

Indexes support searches by capture and timestamp, source IP, destination IP, protocol and destination port. The interface limits the number of rows displayed and supports paging so that large captures do not overwhelm the browser. Database writes are performed in batches inside transactions.

5.2 Questions the tool should answer

- How many packets in this capture use TCP, UDP, ICMP or another protocol?
- Which source and destination addresses appear most often?
- Which packets involve a chosen host or destination port?
- What traffic occurred within a selected time interval?

6 Product Interface

The following screens show how the workflow appears to a user. They use sample values to illustrate the interface and are not capture-specific test evidence.

6.1 Select and validate the capture

The user chooses an authorized PCAP file. The application checks the file type and whether the parser can open it before any packet data is imported.

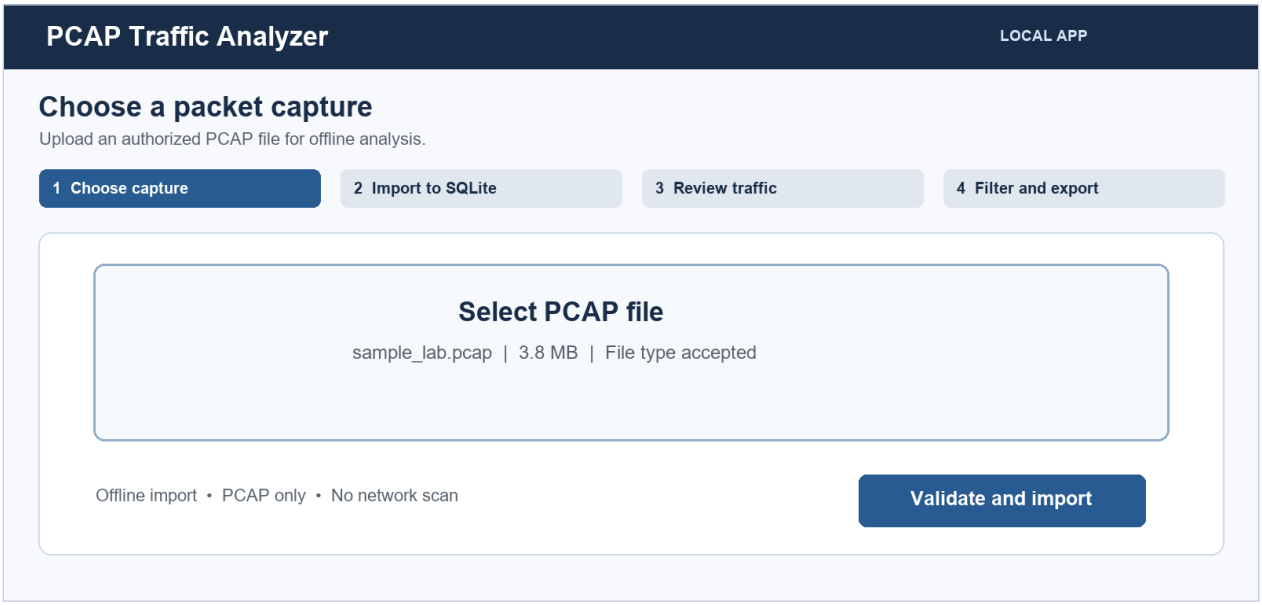


Figure 2: File selection and validation screen (sample values)

6.2 Convert packet data to SQLite

The importer decodes packets, saves normalized metadata and creates a searchable local database. The completion view shows the source file and an import summary.

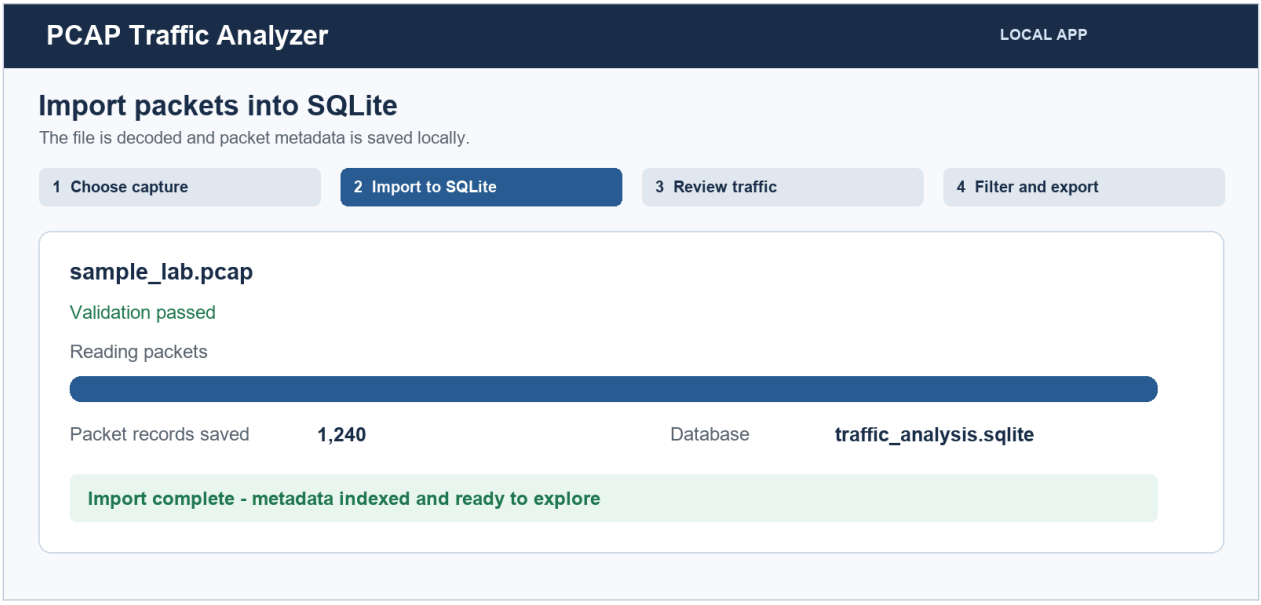


Figure 3: PCAP to SQLite import screen (sample values)

6.3 Review traffic summaries

After import, the dashboard shows packet totals by protocol and the available filters. The packet list keeps capture time, endpoints, protocol and length visible for review.

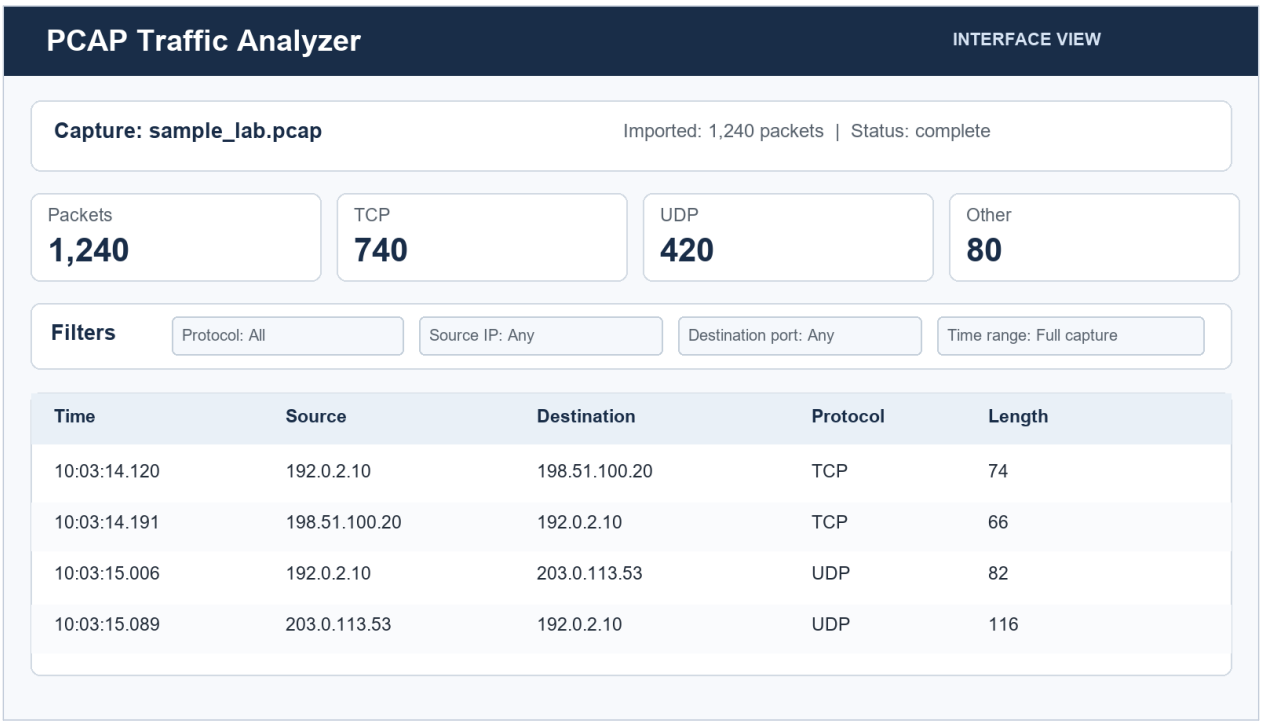


Figure 4: Traffic dashboard (sample values)

6.4 Filter and export packet records

The user can narrow the packet list by protocol, endpoint, destination port or time, then export the selected metadata to CSV.

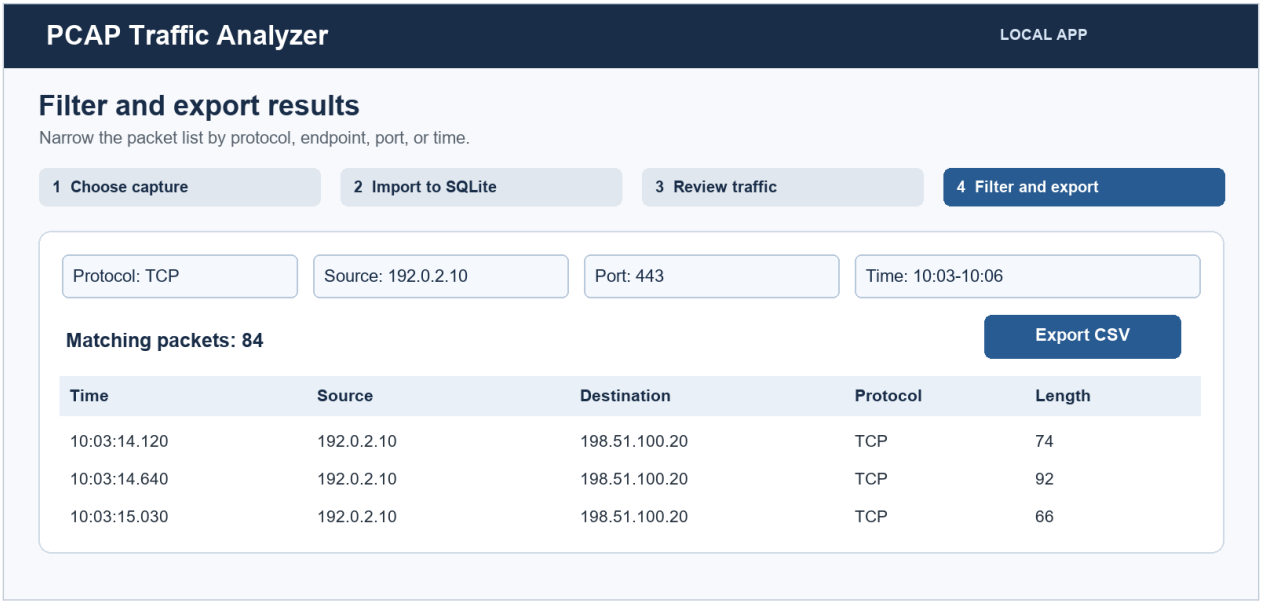


Figure 5: Filtered packet list and CSV export (sample values)

7 Validation Method

Functional checks cover the parser, database, interface and CSV output. Small synthetic captures with known packets provide clear expected values, while authorized classroom captures provide a larger end-to-end check.

ID	Test	Acceptance criterion
T01	Valid PCAP import	Database packet count equals parser count
T02	Known TCP and UDP packets	Protocol, addresses, ports and lengths match expected values
T03	Packet without IP or ports	Import succeeds with null fields where appropriate
T04	Malformed or unsupported file	Clear error and no misleading completed status
T05	Repeat import of same file	Duplicate handling follows the selected policy
T06	Host, protocol, port and time filters	Returned rows match a manually checked fixture
T07	CSV export	Exported rows equal the visible filtered selection
T08	Larger capture	Import finishes within an agreed classroom benchmark

7.1 Test evidence

A reproducible validation record includes the application version, a description of the test capture, the run date, the observed outcomes for T01 to T08 and any unresolved defects. Test captures should be authorized and should exclude personal traffic where possible.

7.2 Acceptance of the required module

The PCAP to SQLite requirement is demonstrated by importing an actual PCAP file, inspecting the packet rows in SQLite and retrieving those rows for filtering or display.

8 Implementation Components

Component	Function	Visible output
File intake	Accept and validate a local PCAP file	Filename and validation message
Parser	Extract timestamps, endpoints, protocol, ports and length	Normalized packet records
SQLite importer	Store packet rows and index searchable fields	Local database and import summary
Analysis queries	Count protocols and filter packet rows	Dashboard totals and result list
CSV export	Write the currently selected metadata rows	Downloadable CSV file

9 Risks and Safeguards

Risk	Safeguard
Sensitive data in captures	Use authorized lab files, avoid storing payloads and restrict local file access
Large file stalls the interface	Set a file size limit, import in batches and page query results
Partial or malformed packets	Allow missing fields, count decode errors and flag incomplete imports
Misleading security claims	Report observations only and avoid automatic attack verdicts
Duplicate or inconsistent data	Record the capture hash and apply a documented re-import policy

10 Conclusion

The PCAP to SQLite Network Traffic Analyzer centers on the required PCAP Analysis module. It turns authorized packet captures into a searchable local dataset and provides basic traffic summaries through a local interface. Its value depends on accurate conversion, clear handling of incomplete data and reproducible validation against known captures.

Appendix A Glossary

Term	Meaning
PCAP	File format for recorded network packets
Packet metadata	Visible fields such as time, addresses, protocol, ports and length
SQLite	Local relational database stored in a single file
Capture	A source file containing recorded packets
Offline analysis	Reading an existing file without collecting live traffic